

CYBER THREAT INTELLIGENCE

RESEARCH REPORT

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SECTION 1: CAMPAIGN RESEARCH — CAMPAIGN C0066

1.1 Campaign Overview & Threat Actor Association

Campaign C0066 represents a sophisticated, multi-stage cyber espionage operation orchestrated by **Earth Lusca** (also tracked as TAG-22, Charcoal Typhoon, and CHROMIUM). This suspected China-based threat group has been active since at least April 2019. The campaign focuses heavily on long-term data exfiltration, credential theft, and establishing persistent infrastructure access within high-value environments.

1.2 Timeline & Objectives

- **Timeline:** Active multi-stage deployments with operations scaling across target landscapes.
- **Objectives:** Systematic collection of strategic intelligence, credential harvesting across critical communications, and establishing unauthorized access channels.

1.3 Targeted Sectors & Countries

The campaign selectively targets organizations of primary geopolitical and economic interest across Asia, Europe, and the Americas, including:

- Government ministries and public infrastructure.
- Media organizations and independent journalists.
- Educational institutions and religious organizations.

1.4 Attack Flow & MITRE ATT&CK Techniques

The campaign relies on exploiting public-facing software vulnerabilities and subverting web trust boundaries.

Tech ID	Tactic	Technique	Description
T1190	Initial Access	Exploit Public-Facing Application	Compromises networks by directly exploiting vulnerabilities in servers like Microsoft Exchange and Oracle GlassFish.
T1566.002	Initial Access	Spearphishing Link	Deploys malicious links to harvest credentials or execute payloads.
T1548.002	Privilege Escalation	Bypass User Account Control	Uses the fodhelper execution pathway to bypass Windows UAC thresholds and achieve administrative tokens.
T1218.005	Defense Evasion	System Binary Proxy Execution	Abuses mshta.exe to load malicious HTML Application (.HTA) scripts inside shortcut (.LNK) components.
T1567	Exfiltration	Exfiltration Over Web Service	Utilizes the legitimate megacmd utility to upload stolen data straight to MEGA cloud storage platforms.

1.5 Real-World Incidents

Earth Lusca has actively manipulated legitimate web assets to inject weaponized JavaScript code for **Watering Hole operations** (T1584.004). They treat compromised servers as an operational launchpad, deploying internal discovery commands such as *ipconfig* (T1016) and running PowerShell utilities like *RDPCConnectionParser* to crawl event logs for administrative patterns.

1.6 Detection Opportunities & Defensive Recommendations

Network Detection: Audit outgoing connections to third-party file storage portals (e.g., MEGA) when invoked by anomalous parent commands. Track incoming URI traffic strings hitting unpatched enterprise applications.

Host Detection: Monitor execution child logs for *mshta.exe* loading non-standard .HTA layers out of shortcut objects.

SECTION 2: SOFTWARE RESEARCH CATALOG (S0660 – S0670)

S0660 **Clambling** (Malware / Modular Backdoor)

A modular backdoor written in C++ utilized by **Threat Group-3390 (APT27)** since at least 2017.

Supported Platforms: Windows

ATT&CK Techniques: T1574.001, T1055.012, T1056.001, T1112, T1082, T1016, T1135, T1083, T1564.001, T1057, T1095, T1566.001, T1027, T1567

Tactic / Metric	Technical Profile & Implementation Details
Execution & Persistence	Drops mpsvc.dll alongside the valid binary MsMpEng.exe to achieve execution via DLL Search Order Hijacking (T1574.001).
Privilege Escalation & Evasion	Executes Process Hollowing (T1055.012) into svchost.exe frames. Can also set local file parameters to hidden states (T1564.001).
Credential Access & Discovery	Runs keyboard capture sequences (T1056.001), captures screen views, and collects hostnames, computer profiles, and Windows builds (T1082). Searches local network paths (T1016), directories (T1083), and network shares (T1135).
C2 & Exfiltration	Communicates over raw TCP/UDP channels (T1095) and exfiltrates files directly to Dropbox services (T1567).

Indicators of Compromise (IoCs)

- Unexpected mpsvc.dll drops in native security paths.
- Rogue memory sections within svchost.exe.
- Outbound data transfers hitting Dropbox IP ranges.

Detection & Mitigation: Validate Microsoft Defender folder modification integrity rules. Monitor non-application network bindings across TCP/UDP layers.

Reference: MITRE ATT&CK S0660 Registry — Clambling

S0661 **FoggyWeb** (Malware / Passive Backdoor)

A highly targeted passive backdoor weaponized by **APT29** since at least early April 2021.

Supported Platforms: Windows Server — Active Directory Federation Services (AD FS)

ATT&CK Techniques: T1574, T1620, T1027.004, T1040, T1550, T1083, T1560.003, T1573, T1041

Tactic / Metric	Technical Profile & Implementation Details
Execution & Persistence	Uses custom loader hooks to inspect CLR's and target execution scopes inside Microsoft.IdentityServer.ServiceHost.exe.
Privilege Escalation & Evasion	Leverages Reflective Code Loading (T1620) to map assembly contents straight to memory without touching disk signatures.

Tactic / Metric	Technical Profile & Implementation Details
Credential Access	Sets up passive listeners to sniff inbound web components (T1040). Harvests token-signing and token-decryption certificates (T1550) to generate fraudulent SAML authorizations.
C2 & Exfiltration	Embeds command tracking codes within legitimate .WebP image streams (T1573). Compresses data via native C# GZipStream routines before exfiltration (T1560.003).

Indicators of Compromise (IoCs)

- Rogue assembly classes running inside the AD FS host process.
- Web logs matching targeted actor URI patterns.
- Unexpected .WebP structures handling high-volume text fields.

Detection & Mitigation: Run deep structural integrity verification checks across AD FS service components. Audit cryptographic token generation pipelines regularly.

Reference: MITRE ATT&CK S0661 Registry — FoggyWeb

S0662 RCTestion (Malware / Backdoor)

A persistent remote access platform written in C++ heavily used by **Mustang Panda** and **TG-3390**.

Supported Platforms: Windows

ATT&CK Techniques: T1574.001, T1055.012, T1056.001, T1113, T1105, T1573, T1082, T1112

Tactic / Metric	Technical Profile & Implementation Details
Execution & Persistence	Utilizes DLL side-loading (T1574.001) to execute inside trusted zones, mapping code segments into hollowed native modules (T1055.012).
Privilege Escalation & Evasion	Spawns a suspended Windows task, clears its memory text, and injects operational instructions into the clean process shell via Process Hollowing (T1055.012).
Credential Access & Discovery	Intercepts live user inputs using an embedded keylogger thread (T1056.001) and takes snapshots of administrative activity using screen captures (T1113).
C2 & Exfiltration	Establishes encrypted outbound communication channels using symmetric cryptography to protect C2 strings (T1573).

Indicators of Compromise (IoCs)

- Unrecognized DLL entries in standard binary paths.
- Unexpected execution changes within native system components.

Detection & Mitigation: Enforce system binary execution tracking. Monitor endpoints for unauthorized, repeated memory hollowing indicators.

Reference: MITRE ATT&CK S0662 Registry — RCTestion

S0663 HUI Loader (Malware / Loader)

A custom security wrapper loader designed to decrypt, unwrap, and execute subsequent spyware arrays like Cobalt Strike or PlugX. Used primarily by **Mustang Panda**.

Supported Platforms: Windows

ATT&CK Techniques: T1574.002, T1027.002, T1140

Tactic / Metric	Technical Profile & Implementation Details
Execution & Persistence	Relies on DLL Side-Loading (T1574.002) by placing malicious companion DLL structures next to legitimate, signed software files.
Privilege Escalation & Evasion	Reads payloads stored as packed, encrypted chunks on disk (T1027.002) and decrypts them directly within memory to evade disk signature monitoring (T1140).

Indicators of Compromise (IoCs)

- Digitally signed software executing out of non-standard locations accompanied by unverified custom libraries.

Detection & Mitigation: Monitor local system directories for newly dropped executable/DLL pairs. Enforce application control policies.

Reference: MITRE ATT&CK S0663 Registry — HUI Loader

S0664 Pandora (Malware / Remote Access Trojan)

An advanced backdoor and system component weaponized in targeted intrusion frameworks to maintain administrative presence.

Supported Platforms: Windows

ATT&CK Techniques: T1055.012, T1112, T1027.015, T1057

Tactic / Metric	Technical Profile & Implementation Details
Execution & Evasion	Injects tracking threads into newly spawned svchost.exe process frames using process hollowing routines (T1055.012).
Persistence & Escalation	Saves setup parameters and structural execution keys as encrypted tokens inside host registry hives (T1112).
Credential Access & Discovery	Continuously queries the OS to map active processes and locate endpoint security agents (T1057).
Command & Control	Uses the QuickLZ compression engine to compress and hide internal code strings from signature scanners (T1027.015).

Indicators of Compromise (IoCs)

- Newly modified binary keys inside system registry paths.

- Active strings indicating QuickLZ runtime decompression loops.

Detection & Mitigation: Deploy system-wide registry monitoring to flag unauthorized persistence edits. Baseline standard memory injection metrics.

Reference: MITRE ATT&CK S0664 Registry — Pandora

S0665ThreatNeedle (Malware / Advanced Backdoor Cluster)

An advanced backdoor cluster belonging to the Manuscript malware family, engineered and deployed by the Lazarus Group.

Supported Platforms: Windows server and workstation architectures

ATT&CK Techniques: T1547.001, T1562.001, T1140, T1204.002

Tactic / Metric	Technical Profile & Implementation Details
Execution & Delivery	Kicks off execution when a user interacts with highly targeted spear-phishing documents via macros (T1204.002).
Persistence	Achieves boot persistence by installing its modules directly into local user %APPDATA% Startup directories (T1547.001).
Defense Evasion	Disables system security configurations and changes registry rules to impair host tools (T1562.001). Decodes runtime parameters via AES/RC4 (T1140).
Lateral Movement	Designed to bridge segmented networks. In air-gapped environments, builds proxy routing paths across infected systems to relay stolen data through internet-facing nodes.

Indicators of Compromise (IoCs)

- Unrecognized files in user startup folders.
- Anomalous internal network routing traffic between isolated server segments.

Detection & Mitigation: Enforce strict internal network micro-segmentation. Audit parent-child execution logs for Microsoft Office applications spawning system tools.

Reference: MITRE ATT&CK S0665 Registry — ThreatNeedle

S0666Gelsemium (Malware / Modular Framework)

A multi-tier malware framework composed of custom droppers, loaders, and primary backdoors built with the Microsoft Foundation Class (MFC) framework.

Supported Platforms: Windows

ATT&CK Techniques: T1055.012, T1083, T1071.001

Tactic / Metric	Technical Profile & Implementation Details
Execution & Evasion	Uses process hollowing (T1055.012) to take over legitimate host applications. Built on MFC to mimic enterprise applications.

Tactic / Metric	Technical Profile & Implementation Details
Discovery	Systematically crawls local filesystems and directory indices to identify targeted intellectual assets (T1083).
Command & Control	Blends into standard web traffic by mimicking regular application layer protocols (T1071.001).

Indicators of Compromise (IoCs)

- Native system tools establishing unexpected outbound web socket paths.
- Signature traces matching MFC application blocks in volatile memory.

Detection & Mitigation: Track and log anomalous outbound web requests originating from native Windows binaries. Run memory-integrity scans to flag hollowing markers.

Reference: MITRE ATT&CK S0666 Registry — Gelsemium

S0667 Chrommme (Malware / Backdoor Tool)

A custom backdoor built with the MFC framework that shares development links and infrastructure overlays with Gelsemium.

Supported Platforms: Windows

ATT&CK Techniques: T1082, T1016, T1033

Tactic / Metric	Technical Profile & Implementation Details
Execution Method	Injected directly into host environments to function as an initial profiling probe with shared framework hooks with Gelsemium.
Discovery & Profiling	Queries system variables to harvest the local computer name (T1082), logs private network adapter IP paths (T1016), and checks system accounts to identify current user permissions (T1033).

Indicators of Compromise (IoCs)

- Rapid execution of system group profiling commands immediately following an unverified software launch.

Detection & Mitigation: Restrict system environment discovery tool access for standard user accounts. Monitor for unusual endpoint network topology mapping activity.

Reference: MITRE ATT&CK S0667 Registry — Chrommme

S0668 TinyTurla (Malware / Lightweight Backdoor)

A minimalist backup backdoor deployed by the **Turla** group as a fail-safe access channel if their primary tools are uncovered.

Supported Platforms: Windows

ATT&CK Techniques: T1543.003, T1071.001, T1027

Tactic / Metric	Technical Profile & Implementation Details
Persistence & Escalation	Registers its execution loop as a background Windows service (T1543.003) to maintain persistence under high-level service execution privileges.
Defense Evasion	Obfuscates internal string arrays and configuration profiles (T1027) to evade static file scanners.
Command & Control	Simulates regular web activity by communicating with C2 infrastructure over HTTPS (T1071.001), waiting silently for secondary instructions.

Indicators of Compromise (IoCs)

- New service registry keys pointing toward hidden or unverified DLL structures.

Detection & Mitigation: Benchmark normal system service registrations. Audit outbound HTTPS connections originating from background service processes.

Reference: MITRE ATT&CK S0668 Registry — TinyTurla

S0669 KOCTOPUS (Malware / Loader)

A multi-tier delivery loader designed to check for system administrative rights and systematically neutralize local defenses.

Supported Platforms: Windows

ATT&CK Techniques: T1548.002, T1562.001, T1105

Tactic / Metric	Technical Profile & Implementation Details
Privilege Escalation	Queries system parameters to execute auto-elevation exploits (eventvwr.exe or sdclt.exe UAC bypass methods) to gain administrative tokens (T1548.002).
Defense Evasion	Locates and deletes configuration keys and registry nodes associated with Microsoft Defender (T1562.001) to clear the path for secondary payloads.
Command & Control	Connects to external servers to pull down secondary ransomware or advanced backdoors after clearing defenses (T1105).

Indicators of Compromise (IoCs)

- Abrupt termination or disabling of security monitoring tools accompanied by registry modifications to Defender paths.

Detection & Mitigation: Enable tamper protection across endpoint defense applications. Restrict non-administrative user permissions from interacting with security registries.

Reference: MITRE ATT&CK S0669 Registry — KOCTOPUS

S0670 WarzoneRAT (Malware / Commercial RAT)

A commercially available Remote Access Trojan sold widely on underground markets, allowing operators to gain full visual and administrative control over endpoints.

Supported Platforms: Windows

ATT&CK Techniques: T1021.001, T1056.002, T1548.002

Tactic / Metric	Technical Profile & Implementation Details
Privilege Escalation	Targets auto-elevating storage manager structures to bypass UAC prompts and grab high-integrity administrative tokens (T1548.002).
Credential Access	Features automated harvesting tools targeting database files inside Google Chrome or Firefox to extract saved usernames and passwords (T1056.002).
Lateral Movement	Can quietly activate host Remote Desktop Protocol (RDP) properties and alter firewall rules, allowing the attacker to establish interactive visual desktop control sessions (T1021.001).

Indicators of Compromise (IoCs)

- Unauthorized out-of-process reads targeting web browser credential stores.
- Sudden RDP initialization requests coming from internal endpoints.

Detection & Mitigation: Restrict cross-workstation internal RDP communication paths. Enforce application control policies to stop unauthorized binary execution.

Reference: MITRE ATT&CK S0670 Registry — WarzoneRAT